Standard Photon–Baryon Transport as the Transfer Interface for OPH Sources

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Abstract

This note shows that standard general-relativistic photon–baryon Boltzmann transport is reproduced on an Observer-Patch Holography (OPH) branch that takes as premises a geometry, photon refinement convergence, an electromagnetic vertex, a polarized collision matrix, the electron mass, and a recombination history; none of these is derived in this note. The reduction gives the linear multipole hierarchy, acoustic and diffusion approximations, and a line-of-sight transfer formula in a common brightness convention; these are the Ma–Bertschinger system and its standard approximations, and nothing in them is derived from observer-patch consistency. The OPH content is two interfaces, a repair-exchange rate and a collar stress, which vanish on the standard branch and mark where a conditional extension would enter.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected codes, radial directions, fields, clocks and couplings may be specified as part of its design; their implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model

in the public repository.[1]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

This paper's hypothesis and place in the tower

This note sits at layer seven as the transfer link between a primordial source and the observed acoustic sky. Its hypothesis is that photon–baryon transport on an OPH branch is the coarse limit of photon-class records continued between commits, with Thomson collisions at commits, so that the branch reduces to the standard Einstein–Boltzmann system plus two OPH interfaces, a repair-exchange rate and a collar stress, which vanish on the standard branch.

The reduction is a conditional theorem. The geometry, the photon refinement convergence condition, the electromagnetic vertex and photon pole, the polarized collision matrix, the electron mass, the recombination history, the FLRW background and the initial spectrum are premises; the acoustic, damping and line-of-sight results are the textbook consequences of the reduced system. The note certifies no initial spectrum, normalization, recombination history or likelihood, and it reads no Planck data.

For the reduction to apply, the executable model must realize on one process a two-polarization photon sector, its continuation between record commits, a physical clock and a smooth Lorentzian background. The native construction supplies none of these [14, 13], and its passive repair has stationary decaying correlations and no wave propagator (Section 6). The reduction is refuted on a realized branch whose photon-class record process has a nonzero scaled quadratic-variation limit, since the transport is then diffusive or jump-like rather than Liouville flow, or whose source-level commit kernel differs from the Thomson functional.

1 Branch inputs

Assumption 1.1 (Einstein branch)**.** *The observer-facing normal form carries a Lorentzian metric g_{ab} satisfying $G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle$, with $G_{\mathrm{geom}} = \ell_\star^2$ from the edge-entropy/area-law theorem and $\Lambda \ell_\star^2 = 3\pi / N_\star$ from capacity closure. This area-law normalization is a premise: a reading of the whole screen's count as the area of a three-dimensional region's boundary requires a region map that the executable construction does not supply [17]. $N_\star$ is the capacity-identity estimate read from measured Λ through a declared capacity-to-scale bridge; Λ is an input to this assumption, not an OPH output. A physical interpretation of $N_\star$ beyond that estimate requires the readback map F and its contraction certificate. The executable carrier bounds dependence by one seam per ordered repair event, with no laboratory speed or continuum light cone [18], and does not supply the smooth metric assumed here. This branch requires physically identified events with exact order reflection or controlled vanishing causal error, count-to-volume convergence, a common clock and physical*

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2058/docs/COMPLETION_PROGRAM.md

stress, together with the capacity premises. Its smooth Einstein reading additionally needs same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar curvature alone is diagnostic.

The flat causal-order, count-volume and count-clock limit of the spacetime construction [9] uses a supplied event law and a relative reference unit; it does not calibrate the common physical clock required by the Einstein, electromagnetic and cosmological transfer assumptions.

Assumption 1.2 (Electromagnetic action and photon pole)**.** *The realized branch carries the Standard Model quotient and the Maxwell kinetic action with a* U(1) *vertex of strength* α*. This note uses* $\alpha = \alpha_{\mathrm{root}}$*, the exact root of a declared source map, conditional on same-scheme transport, target-independent selection of that map, and a typed bridge showing that its two coordinates read the same physical quantity. In addition, this transport note assumes a positive-energy physical QED quantization with two transverse states and a positive-residue isolated pole at* $k^2 = 0$ *on the ordinary Lorentz-vacuum, unbroken electromagnetic phase. This extra photon-pole receipt is not implied by the reconstructed abstract group. Charged-lepton* ratios *are branch outputs; the absolute electron mass* m_e *is empirically anchored. The inputs are the cited electroweak branch theorem, the explicit quantum-pole assumption just stated, and the declared import* m_e*.*

Assumption 1.3 (Cosmological background)**.** *The homogeneous limit of the branch is spatially flat FRW,* $\mathrm{d}s^2 = a^2(\tau)\,[-\mathrm{d}\tau^2 + \delta_{ij}\mathrm{d}x^i\mathrm{d}x^j]$*, with Friedmann dynamics from Assumption 1.1. Scalar perturbations are written in the conformal Newtonian gauge,*

$$\mathrm{d}s^2 = a^2(\tau)\left[-(1+2\psi)\,\mathrm{d}\tau^2 + (1-2\phi)\,\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j\right],$$

with conventions of Ma–Bertschinger (1995). Overdots are ∂_τ*.*

Assumption 1.4 (Recombination import)**.** *The free-electron history* $n_e(\tau)$ *(equivalently* $x_e(\tau)$*) is computed from standard atomic physics (Saha/Peebles-class) using imported atomic data. It is in the same import class as the atomic clock unit of the gravity construction: it contains no CMB-target information, so it is a permissible declared input under the anti-fitting rule, but it is an import, not a branch output.*

Assumption 1.5 (Conditional initial spectrum)**.** *Initial curvature perturbations* $\zeta(\mathbf{k})$ *are Gaussian with spectrum* $\Delta^2_\zeta(k) = A_s\,(k/k_*)^{n_s-1}$*. The Gaussian and statistically isotropic character is the maximum-entropy screen-covariance theorem. A finite tilt certificate requires a source generator, the weighted orientation-half identity, and a physical clock receipt. The radial theorem proves that one-shell inversion is nonunique and supplies physical source-dilation and cross-covariance tomography as conditional uniqueness routes. Finite source instantiation is not supplied here. This note consumes* A_s *and* n_s *as gated inputs and derives no initial spectrum. The registered branch* $n_s = 1 - P/48$ *fixes* P *at its inherited* α*-calibrated value, and the executable observer construction does not derive that branch from its joint records [19].*

2 Conditional Refinement Liouville Transport

Definition 2.1 (One-photon record density)**.** *Conditional on the physical pole in Assumption 1.2, let* $\mathcal{E}_\gamma$ *be its positive-energy photon sector. Between record commits, a single excitation is a record continued across patch overlaps. Coarse-graining patch position and overlap momentum labels defines the phase-space record density* $f(x^i, q, \hat{n}, \tau)$ *on the forward mass shell, where* q *is comoving momentum magnitude and* $\hat{n}$ *the direction.*

Theorem 2.2 (Refinement Liouville)**.** *On a branch with the Maxwell action, quantum photon-pole receipt, and overlap Lorentz kinematics from Assumptions 1.2 and 1.1, let P_r be the quotient-lumpable free-flow transition kernel at step h_r. For every frozen smooth compactly supported test function ϕ, set*

$$L_r\phi = \frac{P_r\phi - \phi}{h_r}, \qquad \Gamma_r(\phi) = \frac{P_r(\phi^2) - (P_r\phi)^2}{h_r}.$$

The photon refinement convergence condition consists of compact containment/tightness, convergence of initial laws, uniform-on-compact generator convergence $L_r\phi \to X_H\phi$ to the null-geodesic Hamiltonian vector field, vanishing quadratic variation $\Gamma_r(\phi) \to 0$, uniqueness of the Hamiltonian ODE, refinement and schedule naturality, and vanishing boundary and phase-space-measure defects. Under this receipt, the between-commit record flow satisfies, in the refinement limit,

$$\frac{Df}{\mathrm{d}\tau} \equiv \frac{\partial f}{\partial \tau} + \frac{\mathrm{d}x^i}{\mathrm{d}\tau}\frac{\partial f}{\partial x^i} + \frac{\mathrm{d}q}{\mathrm{d}\tau}\frac{\partial f}{\partial q} + \frac{\mathrm{d}\hat{n}}{\mathrm{d}\tau}\cdot\frac{\partial f}{\partial \hat{n}} = C[f],$$

where the characteristics are the null geodesics of g_{ab} and $C[f]$ is supported on record commits only. In particular the collisionless flow preserves the Liouville measure.

Proof sketch under the convergence condition. For every test function, the discrete process minus the time integral of $L_r\phi$ is a martingale whose quadratic variation is controlled by $\Gamma_r(\phi)$. Tightness gives subsequential limits and vanishing quadratic variation makes every limiting martingale deterministic. Generator convergence says that each limit solves the martingale problem for X_H, equivalently the null-geodesic Hamiltonian ODE. Uniqueness identifies the whole limit, independently of the accepted repair schedule. Thus $\mathrm{d}x/\mathrm{d}\tau$, $\mathrm{d}q/\mathrm{d}\tau$, and $\mathrm{d}\hat{n}/\mathrm{d}\tau$ are the geodesic-flow coefficients. On the background of Assumption 1.3, $\mathrm{d}q/\mathrm{d}\tau = q[\dot{\phi} - \ldots]$ has the form used in §4.

Measure preservation. The identified limit is Hamiltonian flow on the mass shell, so it preserves the canonical Liouville measure. The finite measure-defect clause checks that the quotient measures converge to that canonical measure rather than to a different coarse volume form. Accepted repair confluence only removes schedule dependence of committed endpoints. It supplies neither an inverse between-commit dynamics nor a phase-space Jacobian. The orientation doubling of a finite label register is bookkeeping and does not produce either property.

Convergence premise. Photon-sector convergence of the finite patch flow to the displayed generator and zero-quadratic-variation limits is not supplied here. The abstract Lean layer proves the corresponding schedule-independence and refinement-naturality statement in its setting. Section 6 gives the corresponding simulator test. Theorem 2.2 is conditional on this convergence premise. □

Remark 2.3 (Scalar continuum on the source family). On the conservative golden-record population of the spacetime construction, a positive source-metric scalar action has a proved finite-time continuum limit. For $m \geq 0$ the finite equation $\ddot{q} + c^2Lq + m^2q = 0$ converges to $u_{tt} - c^2\Delta u + m^2u = 0$ with an explicit detector error bound as $\epsilon \to 0$ and $h/\epsilon^2 \to 0$, and the canonical kick–drift–kick time discretization of the same action converges with a full-spectrum bound. For $m = 0$ the limiting characteristics are the null cone $|dx| = c\,dt$. The read law of the same family has cone radii between $c(1 - 2h/a)\tau$ and $c\tau$, so the two limiting cones coincide when the wave speed of the action is identified with the read-law ratio a/Δ; that identification is a common-model condition and is not a theorem. The high-frequency limit of the massless wave equation is the null-geodesic Liouville flow on the flat mass shell: Wigner measures of its solutions are transported by the Hamiltonian flow of $|\xi|$ (Lions–Paul 1993). On the flat declared family this supplies the free-flow generator X_H of Theorem 2.2 for a scalar record field. It supplies neither the two-polarization photon sector, nor the quotient lumpability and vanishing quadratic variation of the record process, nor the

characteristics of the perturbed metric of Assumption 1.3. The finite Maxwell dynamics on the Whitney cone gives constraint-preserving evolution on a supplied domain and no continuum limit. The photon refinement convergence of Section 6 therefore reduces, on the flat family, to the vector, record-process and curved-background clauses.

Corollary 2.4 (Collision and diffusion alternatives)**.** *If the finite generator decomposes as* $L_r = L_r^{\mathrm{free}} + C_r$ *and* $C_r\phi \to C\phi$*, then the limiting density obeys*

$$\partial_\tau f + \{f, H\} = C^* f.$$

A nonzero limit of Γ_r *excludes the deterministic Liouville conclusion. A diffusion conclusion additionally requires vanishing jumps and a local second-order limiting generator. A jump limit instead contributes its jump operator to the transport equation.*

For example, on two states let $Q\phi(i) = \phi(1-i) - \phi(i)$ and $P_h = I + hQ$, $0 < h \le 1$. For $\phi(i) = i$, the scaled conditional variance is $\Gamma_h(\phi) = 1 - h \to 1$, while the limiting generator Q describes rate-one jumps between the two states. Thus nonzero quadratic variation alone does not identify a diffusion.

Remark 2.5. The transport model places the collision functional at record commits and uses the independently certified measure-preserving flow between them. Claims about entropy production therefore refer to the commit terms counted by $C[f]$. The accepted repair relation and the reversible equilibrium or between-commit dynamics are separate structures.

3 Record Commits and the Thomson Collision Functional

Theorem 3.1 (Scalar-intensity commit-collision functional)**.** *Let commits between photon-class and electron-class records occur with the soft-limit matrix element of the recovered* U(1) *vertex from Assumption 1.2. Then, to first order in perturbations and at low occupancy, the collision functional in Theorem 2.2 is the Thomson functional: in the electron rest frame,*

$$C[f](q,\hat{n}) \;=\; a\, n_e \sigma_T \left[\frac{3}{16\pi}\int \mathrm{d}\Omega'\,(1 + (\hat{n}\cdot\hat{n}')^2)\, f(q,\hat{n}') - f(q,\hat{n})\right], \qquad \sigma_T = \frac{8\pi}{3}\frac{\alpha^2}{m_e^2},$$

boosted to the frame where baryons move with velocity v_b*. The source-derived reversible proposal kernel is assumed to obey detailed balance on the angular state space. Write/check orientation labels alone do not imply that property, and the accepted strict-descent repair relation is not the equilibrium kernel. For every fixed photon energy* q*, isotropic angular distributions are stationary under this Thomson operator. This displayed scalar-intensity kernel does not by itself derive the Stokes-space Thomson matrix needed for polarization.*

Proof sketch. At low occupancy the commit rate between momentum cells is bilinear in the occupancies with kernel given by the squared vertex; the soft (photon energy $\ll m_e$) limit of the recovered U(1) vertex is classical Thomson scattering with the stated total cross-section and $1 + \cos^2$ angular kernel; this step is standard QED evaluated on the branch couplings and is not repeated here. The detailed-balance receipt equates the stationary weights of each reversible proposal pair. The resulting symmetric rest-frame kernel damps angular anisotropy at each fixed q. It does not mix frequencies, so its stationary kernel contains every isotropic spectrum $f(q)$ rather than a unique Bose or Planck occupancy. Momentum conservation per commit, summed over commits, gives equal and opposite momentum transfer densities to the electron–baryon fluid, with coefficient fixed by $\bar{\rho}_\gamma/\bar{\rho}_b$ bookkeeping; this is the Compton drag used in §4. □

Remark 3.2 (Polarization provenance). The polarization hierarchy used below imports the standard polarized Thomson scattering matrix acting on Stokes brightness. The scalar $1 + \cos^2\theta$ kernel displayed in Theorem 3.1 is its intensity projection, not a derivation of the Q/U collision terms. A source-level OPH derivation of the polarized commit kernel from the recovered vertex is not supplied here. The polarization sector and the associated 16/15 diffusion factor are textbook imports rather than consequences of the displayed operator.

Remark 3.3 (Blackbody provenance). The Planck background spectrum is an initial-condition and thermal-history import in this note, not a consequence of the displayed Thomson functional. Even recoil-inclusive Comptonization conserves photon number and generically approaches a Bose–Einstein spectrum with a chemical potential. Selecting the zero-chemical-potential Planck spectrum requires the appropriate early thermalization history, including photon-number-changing channels such as double Compton scattering and bremsstrahlung, or an independently justified blackbody initial condition. An OPH derivation of the blackbody spectrum requires those channels and their rates rather than inferring spectral uniqueness from angular detailed balance.

Remark 3.4 (Provenance of the normalization). α_{root} is an exact declared-map root. Its use as the physical vertex strength is conditional on the three bridge premises in Assumption 1.2. m_e is an import (Assumption 1.2): OPH supplies no source-derived charged-lepton absolute mass, so σ_T's normalization is an explicit import. The recombination history $n_e(\tau)$ is imported under Assumption 1.4.

Remark 3.5 (OPH-specific channels). The declared two-channel continuation permits the following modifications at linear order: (i) a number-changing repair-exchange channel $\Gamma_{\mathrm{rec}}(k,a)$ adds a monopole source and a spectral-distortion term; it enters $C[f]$ additively and requires an active fiber, a physical clock, and a common-parent response; (ii) a collar-packet stress $\rho_A, B_A(k,a)$ enters only the gravitational source side (the Einstein constraints in §4), never $C[f]$. Setting both declared corrections to zero recovers standard transport.

4 The Linear Multipole Hierarchy

Theorem 4.1 (Reduction to the standard hierarchy)**.** *Linearize Theorems 2.2–3.1 on the background of Assumption 1.3 with scalar perturbations, expand the fractional intensity brightness $F_\gamma(\mathbf{k},\hat{n},\tau)$ and polarization brightness G_γ as $F_\gamma = \sum_\ell (-i)^\ell (2\ell+1) F_{\gamma\ell} P_\ell(\mu)$, and likewise for G_γ, with $\mu = \hat{k}\cdot\hat{n}$. Writing $\delta_\gamma = F_{\gamma 0}$, $\theta_\gamma = \frac{3}{4}kF_{\gamma 1}$, $\sigma_\gamma = \frac{1}{2}F_{\gamma 2}$, and $\dot\kappa \equiv a n_e \sigma_T$, the system is (Ma–Bertschinger 1995, eq. 64):*

$$
\begin{aligned}
\dot\delta_\gamma &= -\tfrac{4}{3}\theta_\gamma + 4\dot\phi,\\
\dot\theta_\gamma &= k^2\left(\tfrac{1}{4}\delta_\gamma - \sigma_\gamma\right) + k^2\psi + \dot\kappa\,(\theta_b - \theta_\gamma),\\
2\dot\sigma_\gamma &= \tfrac{8}{15}\theta_\gamma - \tfrac{3}{5}kF_{\gamma 3} - \tfrac{9}{5}\dot\kappa\,\sigma_\gamma + \tfrac{1}{10}\dot\kappa\,(G_{\gamma 0} + G_{\gamma 2}),\\
\dot F_{\gamma\ell} &= \frac{k}{2\ell+1}\left[\ell F_{\gamma(\ell-1)} - (\ell+1)F_{\gamma(\ell+1)}\right] - \dot\kappa F_{\gamma\ell}, \qquad \ell \geq 3,\\
\dot G_{\gamma\ell} &= \frac{k}{2\ell+1}\left[\ell G_{\gamma(\ell-1)} - (\ell+1)G_{\gamma(\ell+1)}\right] + \dot\kappa\left[-G_{\gamma\ell} + \tfrac{1}{2}\left(F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2}\right)\left(\delta_{\ell 0} + \tfrac{\delta_{\ell 2}}{5}\right)\right],
\end{aligned}
$$

coupled to baryons,

$$
\dot\delta_b = -\theta_b + 3\dot\phi, \qquad \dot\theta_b = -\frac{\dot a}{a}\theta_b + c_b^2 k^2 \delta_b + k^2\psi + \frac{4\bar\rho_\gamma}{3\bar\rho_b}\,\dot\kappa\,(\theta_\gamma - \theta_b),
$$

and to the Einstein constraints of Assumption 1.1,

$$k^2\phi + 3\frac{\dot a}{a}\left(\dot\phi + \frac{\dot a}{a}\psi\right) = -4\pi G a^2\,\delta\rho_{\rm tot}, \qquad k^2(\phi-\psi) = 12\pi G a^2(\bar\rho+\bar p)\sigma_{\rm tot},$$

where $\delta\rho_{\rm tot}$ and $\sigma_{\rm tot}$ include the gated ρ_A, B_A contributions of Remark 3.5. These contributions are zero on the receipt-failing branch. With $\Gamma_{\rm rec} = B_A = 0$ this is verbatim the standard Einstein–Boltzmann system.

Proof. For scalar intensity, insert the geodesic characteristics of Theorem 2.2 on the perturbed metric, the collision functional of Theorem 3.1, keep first order, and project on Legendre multipoles. The $G_{\gamma\ell}$ equations additionally use the imported polarized Thomson matrix stated above. The subsequent calculation is the textbook computation (Ma–Bertschinger 1995; Dodelson, *Modern Cosmology*). The two conditional OPH channels appear at their unique insertion points. □

5 Acoustic normal form, damping, and the line of sight

Proposition 5.1 (Acoustic oscillations)**.** *In the tight-coupling regime $\dot\kappa \gg k, \frac{\dot a}{a}$, neglect baryon thermal pressure and define $R \equiv 3\bar\rho_b/4\bar\rho_\gamma$ and $\Theta_0 = \delta_\gamma/4$. To leading order in $1/\dot\kappa$,*

$$\frac{\mathrm d}{\mathrm d\tau}\Big[(1+R)\,\dot\Theta_0\Big] + \frac{k^2}{3}\,\Theta_0 = -\frac{k^2}{3}(1+R)\,\psi + \frac{\mathrm d}{\mathrm d\tau}\Big[(1+R)\,\dot\phi\Big],$$

a driven oscillator with sound speed $c_s^2 = 1/[3(1+R)]$. Its homogeneous WKB phase, when the coefficients vary slowly compared with an oscillation, is kr_s, where

$$r_s(\tau_*) = \int_0^{\tau_*} c_s\,\mathrm d\tau,$$

at last scattering τ_*. A cosine initial phase gives the approximate unforced extrema $k_n r_s \simeq n\pi$. Baryon loading changes the equilibrium offset and relative compression and rarefaction amplitudes; evolving gravitational potentials drive the oscillator. Initial phases, free-streaming species, finite visibility and angular projection shift the observed peak positions and heights. The oscillator therefore identifies the acoustic mechanism without fixing exact peaks of a measured TT spectrum.

Proposition 5.2 (Silk damping)**.** *At next order in $1/\dot\kappa$ the oscillator acquires the diffusion envelope e^{-k^2/k_D^2} with*

$$k_D^{-2} = \int_0^\tau \frac{\mathrm d\tau'}{6(1+R)\,\dot\kappa}\left[\frac{R^2}{1+R} + \frac{16}{15}\right],$$

the 16/15 arising from the polarization terms of Theorem 4.1; photon diffusion between commits is the physical mechanism, and its OPH reading is transparent: on scales crossed by a random walk of commit-free transport legs, the record field cannot maintain contrast.

Proposition 5.3 (Line-of-sight solution)**.** *For $k > 0$, with visibility $g(\tau) = \dot\kappa\, e^{-\kappa(\tau,\tau_0)}$ and an optically thick initial endpoint, the temperature multipoles today, for $\ell \geq 2$, are*

$$\Theta_\ell(k,\tau_0) = \int_0^{\tau_0}\mathrm d\tau\,\Big\{g\Big[\Theta_0 + \psi + \tfrac{\Pi}{4}\Big]\,j_\ell(k(\tau_0-\tau)) + \Big(\frac{g\,v_b}{k}\Big)^{\cdot} j_\ell + \frac{3}{4k^2}(g\Pi)^{\cdot\cdot} j_\ell + e^{-\kappa}(\dot\psi+\dot\phi)\,j_\ell\Big\},$$

($\Theta_\ell = F_{\gamma\ell}/4$, $\Pi = (F_{\gamma 2} + G_{\gamma 0} + G_{\gamma 2})/4$, $v_b = \theta_b/k$; every unlabelled j_ℓ has argument $k(\tau_0-\tau)$), and

$$C_\ell^{TT} = 4\pi\int\frac{\mathrm dk}{k}\,\Delta_\zeta^2(k)\,\left|\frac{\Theta_\ell(k,\tau_0)}{\zeta(k)}\right|^2.$$

Given the conditional (A_s, n_s) of Assumption 1.5 and the imports of Assumptions 1.2–1.4, this is the standard linear transfer formula. Agreement of an independent implementation with CAMB at identical inputs and sub-percent tolerances is a separate numerical acceptance test.

The normalization can be checked directly from the stated hierarchy. Dividing the intensity collision term of Ma–Bertschinger by four gives $-\dot\kappa\Pi P_2(\mu)/2$ and the Doppler term $-i\mu\dot\kappa\theta_b/k$. Along the line of sight, $P_2(\mu) = (3\mu^2-1)/2$ and $\partial_\tau e^{ik\mu(\tau-\tau_0)} = ik\mu e^{ik\mu(\tau-\tau_0)}$. Integration by parts therefore gives $g\Pi/4 + 3(g\Pi)^{\cdot\cdot}/(4k^2)$ and $(g\theta_b)^{\cdot}/k^2$, respectively. The initial boundary terms vanish by optical thickness, and the observer terms affect only the monopole and dipole. These conventions specify the signs and factors in the Seljak–Zaldarriaga line-of-sight construction.

6 Premises and Numerical Tests

Condition	Statement	Classification
Photon refinement convergence	Finite-lattice refinement Liouville: coarse-grained photon-class record transport has quotient lumpability, tightness, convergent initial laws and first conditional moments, vanishing scaled second conditional moments, generator convergence on a frozen test basis, unique geodesic characteristics, and vanishing schedule, refinement, boundary, mass-shell, and measure defects. This condition is independent of repair confluence and of the orientation-doubled label count.	scalar free-flow generator supplied on the flat declared family (Remark 2.3); vector, record-process and curved-background clauses: simulator test below
Thomson normalization	σ_T normalization: absolute m_e is a declared import; the charged-family absolute-scale theorem is not supplied.	declared import
Recombination history	$x_e(\tau)$ recombination history from imported atomic data; contains no CMB-target information.	declared import
Numerical transfer test	Independent hierarchy implementation matches CAMB temperature and polarization spectra at stated tolerances with identical inputs.	not supplied; engineering test specified below
Tilt certificate	Conditional $n_s = 1 - P_\star/48$; source generator, weighted half, and clock normalization are not supplied.	not supplied
Amplitude + lift	A_s and radial screen-to-k map.	conditional radial theorem; finite source receipt not supplied
Kernels	Γ_{rec}, ρ_A, B_A physical receipts.	not supplied
Likelihood	Official likelihood, provenance, and data-use contract.	not supplied

Simulator test for photon refinement convergence. Freeze a smooth compactly supported test basis. At each regulator estimate the first conditional moment $\widehat{L}_r\phi$ and scaled conditional variance $\widehat{\Gamma}_r(\phi)$, together with quotient-measure, mass-shell, schedule, refinement, boundary-leakage,

and collision/free-channel residuals. Pass requires $\widehat{L}_r\phi \to X_H\phi$ and $\widehat{\Gamma}_r(\phi) \to 0$ on the frozen basis with the preregistered uniform compact envelopes. Random-schedule tests are supplemental. A nonzero second-moment limit excludes the deterministic Liouville theorem; the limiting jump sizes and generator distinguish diffusion from jump transport.

Native repair diffusion and the photon interface. For the specified fair conservative port repair, mean load evolves as $e^{-tL/2}$, with t the native attempt count divided by the seam count, and stationary density has covariance $\kappa e^{-|t-s|L/2}\Pi$. Its integrated readback has a fixed-graph Gaussian limit with inverse-Laplacian covariance; the literal finite-time average has the different normalization $\kappa h(\lambda T/2)$. The exact derivation, local attempted-event record correction and finite-state controls are given in the screen-microphysics companion [10] and preserved in the observer-dynamics evidence archive [20]. Along one common subsequence of refining source screens, actual native density snapshots have joint Gaussian limits with covariance $\exp(-|\tau - u|A)$, where τ is the native attempt count divided by $1260 \cdot 16^l$ and A is a positive self-adjoint operator of heat-trace spectral dimension two on the screen [16, 15]. Coherent waves $\cos(t\sqrt{A})$ require a declared wave processor whose parameter differs from passive waiting, and they propagate at most at speed $1/\sqrt{2}$ in the flat comparison distance per coherent wave parameter [14]. This two-dimensional screen operator supplies no $3+1$ null-geodesic flow or photon sector, and its density limit is a stationary Gaussian process, not a deterministic Liouville flow. These statements instantiate a diffusion/record mechanism. They do not verify the vanishing conditional second moment required for a deterministic photon Liouville limit, nor do they provide polarization, Thomson normalization or recombination.

A nondecreasing global change of time keeps every positive mean mode $\exp[-\lambda g(t)/2]$ positive and decreasing. It cannot supply an oscillatory wave propagator. For any current-state centered observable of a stationary reversible continuous-time repair chain, its autocorrelation is a nonnegative sum of decaying exponentials. The corresponding discrete-time conclusion requires a nonnegative spectrum, as in a lazy chain; a nonlazy two-state flip is a counterexample. These restrictions apply to the stated generator and readout, not to extended dynamics with momentum, active forcing, nonstationary preparation or different records. The passive source matching and acoustic initial data used by a conditional CMB transport calculation are additional physical inputs.

7 Scope of the Conditional Reduction

The cited branch theorems supply the *scalar angular form* of the transport and locate the two conditional OPH kernels. The polarization hierarchy, blackbody background, absolute normalizations, and recombination history are imported. A physical cosmic microwave background prediction requires a source-level polarized collision bridge, blackbody and thermalization provenance, photon refinement convergence, the numerical transfer test, a physical radial lift, a finite primordial source, active-kernel evidence, and an immutable likelihood contract. This reduction certifies no initial conditions, normalizations, recombination history, or likelihood, and it reads no Planck data. Under the anti-fitting rule, the acoustic peak structure is attributable to standard plasma physics. OPH's falsifiable exposure is concentrated in the primordial inputs.

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